

Delta Lake Incremental Data Processing Assignment Report

Objective

The objective of this assignment was to perform incremental data processing using Delta Lake. The tasks included loading a dataset into a Delta table, cleaning the data, processing incremental records using the MERGE operation, and validating the final output.

Dataset

Two datasets were used in this assignment:

- **Dataset_1.csv** – Contains the initial customer data with some duplicate records and null values.
- **Dataset_2.csv** – Contains new customer records and updates for existing customers.

The dataset contains customer details such as customer ID, first name, last name, gender, age, city, state, email, join date, and membership.

Steps Performed

1. Data Loading

The master dataset was uploaded to Databricks and loaded into a Spark DataFrame using PySpark.

2. Data Cleaning

Before processing the data, basic cleaning was performed.

- Removed records containing null values using `dropna()`.
- Removed duplicate records using `dropDuplicates()`.

After cleaning, the dataset contained **43 records**.

3. Creating a Delta Table

The cleaned dataset was saved as a Delta table named **customer_delta** using the Delta format.

4. Incremental Data Processing

The incremental dataset was loaded into Databricks. A Delta Lake **MERGE** operation was performed to process the incremental data.

The MERGE operation:

- Updated existing customer records based on **customer_id**.
- Inserted new customer records that were not present in the Delta table.

5. Validation

After the MERGE operation, the final Delta table was checked to ensure the data was updated correctly.

Validation included:

- Checking the final row count.

- Verifying that duplicate customer IDs were not present.
- Displaying the final dataset.

The final Delta table contained **51 records**.

Result

The assignment was completed successfully. The Delta Lake MERGE operation updated existing customer records and inserted new customer records into the Delta table. The data cleaning process removed null values and duplicate records, and the final dataset was validated successfully.

Conclusion

This assignment helped me understand how Delta Lake supports incremental data processing. I learned how to load data into a Delta table, clean data, perform MERGE operations, and validate the final output using Databricks and PySpark.